

Power Series solutions of DEs

6.2 Solutions About ordinary Points

Consider the DE :

$$A_2(x) y'' + A_1(x) y' + A_0(x) y = 0.$$

We can divide it by $A_2(x)$ to get the standard form equation :

$$y'' + P(x) y' + Q(x) y = 0.$$

Definition A point x_0 is an ordinary point of the DE if both $P(x)$ and $Q(x)$ are analytic at x_0 .

Otherwise, x_0 is a singular Point of the DE.

ex: ① Is $x_0 = 0$ an ordinary point

of the DE $y'' + e^x y' + x^3 y = 0$?

$$\begin{cases} P(x) = e^x = 1 + x + \frac{x^2}{2!} + \dots & \text{for all } x \\ Q(x) = x^3 \end{cases}$$

$P(x)$ and $Q(x)$ are analytic at $x_0 = 0$,

then $x_0 = 0$ is an ordinary point of the DE.

ex: ② Is $x_0 = 0$ an ordinary point of

the DE $y'' + \frac{1}{x} y' + 3y = 0$?

$P(x) = 1/x$ is not analytic at $x_0 = 0$

$Q(x) = 3$ is analytic at $x_0 = 0$

Because of $P(x)$, $x_0 = 0$ is a

Singular point of the DE.

ex: ③ Find all singular points of
the DE $(x^2 - 9)y'' + 2xy' + 5y = 0$.

Standard form

$$y'' + \frac{2x}{x^2 - 9} y' + \frac{5}{x^2 - 9} y = 0$$

$$\left. \begin{aligned} P(x) &= \frac{2x}{x^2 - 9} \\ Q(x) &= \frac{5}{x^2 - 9} \end{aligned} \right\} \begin{array}{l} \text{They are not analytic} \\ \text{at } x = \pm 3. \end{array}$$

$\Rightarrow$ $x = \pm 3$ are the
singular points of the DE.

Theorem

Existence of Power series Solutions

If x_0 is an ordinary point of the DE

$$A_2(x) y'' + A_1(x) y' + A_0(x) y = 0$$

then we can always find two linearly independent power series solutions centered at x_0

$$y = \sum_{n=0}^{\infty} a_n (x - x_0)^n$$

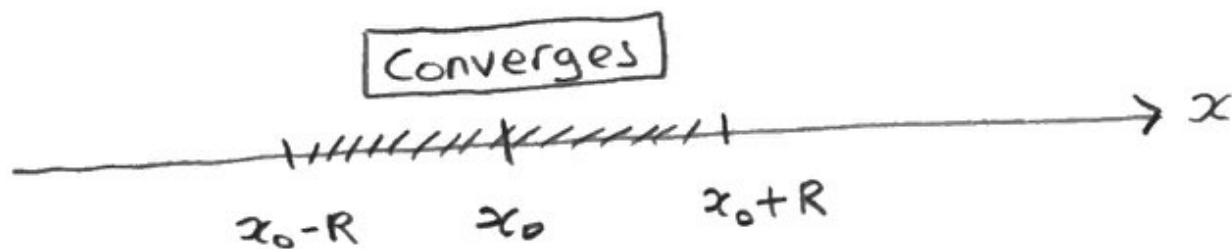
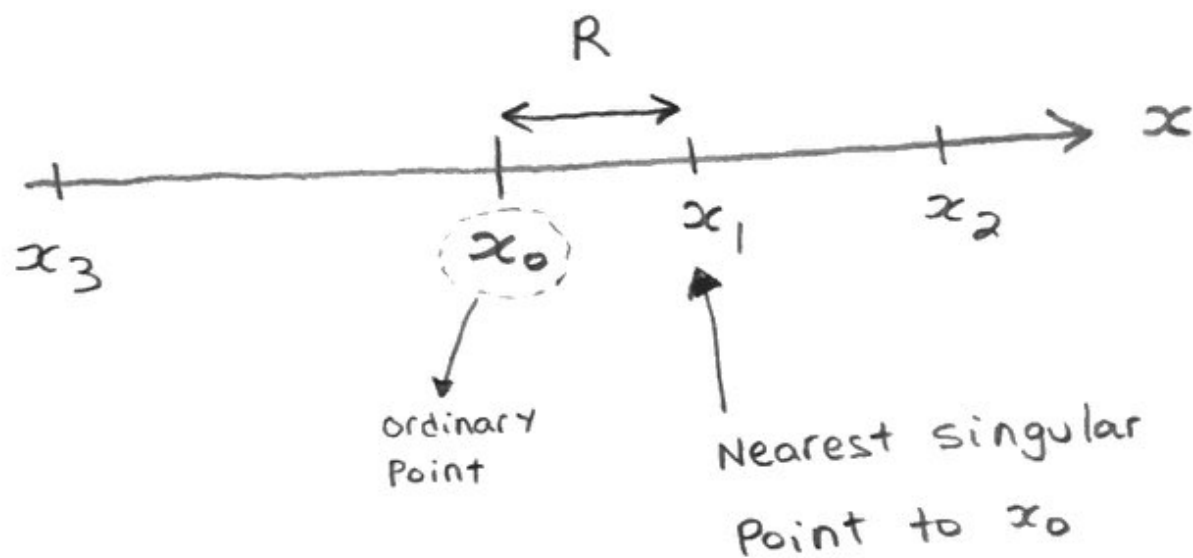
$$= a_0 + a_1(x - x_0) + a_2(x - x_0)^2 + \dots$$

The power series converges at least

for $|x - x_0| < R$ where

R is the distance from x_0 to the closest singular point of the DE.

Suppose that x_1, x_2, x_3 are
singular points.



The series would converge at least
over :

$$x_0 - R < x < x_0 + R$$

If there are no singular point, then $R = +\infty$
and the series converges for all x .

In the examples that we will do,
for the sake of simplicity, we will always
find power series solutions centered
at the ordinary point $x_0 = 0$.

The power series will have the form:

$$y = \sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + \dots$$

We can compute derivatives y' and y''
termwise:

$$\begin{cases} y' = \sum_{n=1}^{\infty} n a_n x^{n-1} = a_1 + 2a_2 x + 3a_3 x^2 + \dots \\ y'' = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} = 2a_2 + 6a_3 x + \dots \end{cases}$$

ex: ④ Solve $y'' + 3x^2 y' - 2y = 0$.

Since $P(x) = 3x^2$ and $Q(x) = -2$ are both analytic at $x_0 = 0$, then $x_0 = 0$ is an ordinary point of the DE.

We can look for power series solutions of the form: $y = \sum_{n=0}^{\infty} a_n x^n$.

Since the DE has no singular point, the series solutions will converge for all x .

$$y = \sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \dots$$

$$y' = \sum_{n=1}^{\infty} n a_n x^{n-1} = a_1 + 2a_2 x + 3a_3 x^2 + \dots$$

$$y'' = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} = 2a_2 + 6a_3 x + \dots$$

Plug the series for y, y', y'' in the DE.

$$y'' + 3x^2 y' - 2y = 0$$

$$\sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} + 3x^2 \sum_{n=1}^{\infty} n a_n x^{n-1} - 2 \sum_{n=0}^{\infty} a_n x^n = 0$$

Note: $3x^2 \sum_{n=1}^{\infty} n a_n x^{n-1} = \sum_{n=1}^{\infty} 3n a_n x^{n+1}$

Distribute the $3x^2$ to the terms of the series

$$\underbrace{\sum_{n=2}^{\infty} n(n-1)a_n x^{n-2}}_{\substack{k=n-2 \rightarrow n=k+2 \\ n=2 \rightarrow k=0}} + \underbrace{\sum_{n=1}^{\infty} 3n a_n x^{n+1}}_{\substack{k=n+1 \rightarrow n=k-1 \\ n=1 \rightarrow k=2}} - \underbrace{\sum_{n=0}^{\infty} 2a_n x^n}_{k=n} = 0$$

$$\sum_{k=0}^{\infty} (k+2)(k+1)a_{k+2} x^k + \sum_{k=2}^{\infty} 3(k-1)a_{k-1} x^k - \sum_{k=0}^{\infty} 2a_k x^k = 0$$

Observe that:

$$\sum_{k=0}^{\infty} (k+2)(k+1)a_{k+2}x^k = \underbrace{2a_2}_{k=0} + \underbrace{6a_3x}_{k=1} + \sum_{k=2}^{\infty} (k+2)(k+1)a_{k+2}x^k$$

$$\sum_{k=0}^{\infty} 2a_kx^k = \underbrace{2a_0}_{k=0} + \underbrace{2a_1x}_{k=1} + \sum_{k=2}^{\infty} 2a_kx^k$$

We can combine all the series as a single one.

$$\underbrace{(2a_2 - 2a_0)}_{k=0} + \underbrace{(6a_3 - 2a_1)x}_{k=1} + \sum_{k=2}^{\infty} [(k+2)(k+1)a_{k+2} + 3(k-1)a_{k-1} \dots - 2a_k] x^k = \textcircled{0}$$

This zero means 0 for all x .

You can think of it as: $0 = \underline{0} + \underline{0}x + \underline{0}x^2 + \underline{0}x^3 + \dots$

all coefficients of x^k are 0

Next, we set all coefficients of x^k equal to zero. We are matching coefficients on both sides.

$$\begin{cases} 2a_2 - 2a_0 = 0 & (k=0) \\ 6a_3 - 2a_1 = 0 & (k=1) \\ (k+2)(k+1)a_{k+2} + 3(k-1)a_{k-1} - 2a_k = 0 & (k=2, 3, 4, \dots) \end{cases}$$

Solve for the a_k of higher subscript in each equation to obtain a recurrence relation.

$$\Rightarrow \begin{cases} a_2 = a_0 \\ a_3 = a_1/3 \\ a_{k+2} = \frac{2a_k - 3(k-1)a_{k-1}}{(k+2)(k+1)} \end{cases}$$

Recurrence relation

for $k=2, 3, 4, \dots$

Starting from a_0 and a_1 , we
can iterate to compute: $a_2, a_3, a_4, \dots$

We need two Linearly independent
Power series solutions y_1 and y_2 .

To get y_1 , start from $a_0 = 1, a_1 = 0$

To get y_2 , start from $a_0 = 0, a_1 = 1$

once we have: $a_0, a_1, a_2, a_3, \dots$

Each series will be:

$$a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \dots$$

y_1	y_2
$a_0 = 1$ $a_1 = 0$	$a_0 = 0$ $a_1 = 1$
$a_2 = a_0 = 1$ $a_3 = a_1/3 = 0$	$a_2 = a_0 = 0$ $a_3 = a_1/3 = 1/3$
$\boxed{k=2}$ $a_4 = \frac{2a_2 - 3a_1}{12} = 1/6$	$a_4 = \frac{2a_2 - 3a_1}{12} = -1/4$
$\boxed{k=3}$ $a_5 = \frac{2a_3 - 6a_2}{20} = -3/10$	$a_5 = \frac{2a_3 - 6a_2}{20} = 1/30$
etc...	etc...

$$\begin{cases} y_1 = 1 + x^2 + \frac{1}{6}x^4 - \frac{3}{10}x^5 + \dots \\ y_2 = x + \frac{1}{3}x^3 - \frac{1}{4}x^4 + \frac{1}{30}x^5 + \dots \end{cases}$$

The general solution of the DE is

$$y = c_1 y_1 + c_2 y_2$$

$$y = c_1 \left(1 + x^2 + \frac{1}{6} x^4 - \frac{3}{10} x^5 + \dots \right) + c_2 \left(x + \frac{1}{2} x^3 - \frac{1}{4} x^4 + \frac{1}{30} x^5 + \dots \right)$$

$$\text{If } y = a_0 + a_1 x + a_2 x^2 + \dots$$

$$\text{then } y' = a_1 + 2a_2 x + \dots$$

$$\text{observe that: } y(0) = a_0 \text{ and } y'(0) = a_1$$

If we have an IVP, start from:

$$\begin{cases} a_0 = y(0) \\ a_1 = y'(0) \end{cases}$$

and iterate to get: $a_2, a_3, a_4, \dots$

ex: ⑤ Find a power series solution of

the IVP:
$$\begin{cases} y'' + 2xy' + 2y = 0 \\ y(0) = 2 \\ y'(0) = -5 \end{cases}$$

Find the first six nonzero terms of the series.

$x_0 = 0$ is an ordinary point of the DE.

Let $y = \sum_{n=0}^{\infty} a_n x^n$

$$\Rightarrow y' = \sum_{n=1}^{\infty} n a_n x^{n-1}$$

$$\Rightarrow y'' = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2}$$

Plug y , y' , and y'' in the DE.

$$\sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} + 2x \sum_{n=1}^{\infty} na_n x^{n-1} + 2 \sum_{n=0}^{\infty} a_n x^n = 0$$

$$\underbrace{\sum_{n=2}^{\infty} n(n-1)a_n x^{n-2}}_{\substack{k=n-2 \rightarrow n=k+2 \\ n=2 \rightarrow k=0}} + \underbrace{\sum_{n=1}^{\infty} 2na_n x^n}_{k=n} + \underbrace{\sum_{n=0}^{\infty} 2a_n x^n}_{k=n} = 0$$

$$\sum_{k=0}^{\infty} (k+2)(k+1)a_{k+2} x^k + \sum_{k=1}^{\infty} 2ka_k x^k + \sum_{k=0}^{\infty} 2a_k x^k = 0$$

Write explicitly the $k=0$ terms of the 1st and 3rd series and combine the series.

$$\underbrace{(2a_2 + 2a_0)}_{k=0} + \sum_{k=1}^{\infty} ((k+2)(k+1)a_{k+2} + 2ka_k + 2a_k) x^k = 0$$

Set the coefficients of x^k equal to 0.

$$\Rightarrow \begin{cases} 2a_2 + 2a_0 = 0 \\ (k+2)(k+1)a_{k+2} + 2ka_k + 2a_k = 0 \\ \text{for } k=1, 2, 3, \dots \end{cases}$$

$$\Rightarrow \begin{cases} a_2 = -a_0 \\ a_{k+2} = \frac{-2ka_k - 2a_k}{(k+2)(k+1)} = \frac{-2a_k(\cancel{k+1})}{(k+2)(\cancel{k+1})} \end{cases}$$

$$\Rightarrow \begin{cases} a_2 = -a_0 \\ a_{k+2} = \frac{-2a_k}{k+2} \text{ for } k=1, 2, 3, \dots \end{cases}$$

The initial conditions give the values of a_0 and a_1 .

$$\begin{cases} a_0 = y(0) = 2 \\ a_1 = y'(0) = -5 \end{cases}$$

$$a_2 = -a_0 = -2$$

$$(k=1) \quad a_3 = -\frac{2a_1}{3} = 10/3$$

$$(k=2) \quad a_4 = -\frac{2a_2}{4} = 1$$

$$(k=3) \quad a_5 = -\frac{2a_3}{5} = -4/3$$

etc...

The solution of the IVP is

$$y = 2 - 5x - 2x^2 + \frac{10}{3}x^3 + x^4 - \frac{4}{3}x^5 + \dots$$

Since the DE $y'' + 2xy' + 2y = 0$

has no singular point, the series

Solution :

$$y = 2 - 5x - 2x^2 + \frac{10}{3}x^3 + x^4 - \frac{4}{3}x^5 + \dots$$

Converges for all x .

Summary

- ① If you don't have an IVP, you need two LI solutions y_1 and y_2 .

Iterate from: $\begin{cases} a_0 = 1, a_1 = 0 & \text{to get } y_1 \\ a_0 = 0, a_1 = 1 & \text{to get } y_2 \end{cases}$

- ② If you have an IVP, iterate from:

$$\begin{cases} a_0 = y(0) \\ a_1 = y'(0) \end{cases}$$

ex: ⑥ Use Power series to solve

$$(x+2)y'' - y = 0.$$

Expand to get:

$$xy'' + 2y'' - y = 0.$$

Let $y = \sum_{n=0}^{\infty} a_n x^n$

$$\Rightarrow y' = \sum_{n=1}^{\infty} n a_n x^{n-1}$$

$$\Rightarrow y'' = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2}$$

Plug y, y', y'' in the DE.

$$x \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} + 2 \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} - \sum_{n=0}^{\infty} a_n x^n = 0$$

$$\underbrace{\sum_{n=2}^{\infty} n(n-1) a_n x^{n-1}}_{\substack{K=n-1 \rightarrow n=K+1 \\ n=2 \rightarrow K=1}} + \underbrace{\sum_{n=2}^{\infty} 2n(n-1) a_n x^{n-2}}_{\substack{K=n-2 \rightarrow n=K+2 \\ n=2 \rightarrow K=0}} - \underbrace{\sum_{n=0}^{\infty} a_n x^n}_{K=n} = 0$$

$$\sum_{K=1}^{\infty} (K+1)K a_{K+1} x^K + \sum_{K=0}^{\infty} 2(K+2)(K+1) a_{K+2} x^K - \sum_{K=0}^{\infty} a_K x^K = 0$$

Write explicitly the $K=0$ terms and
Combine the series.

$$\underbrace{(4a_2 - a_0)}_{\substack{K=0 \text{ of the} \\ 2^{\text{nd}} \text{ and } 3^{\text{rd}} \\ \text{series}}} + \sum_{K=1}^{\infty} [(K+1)K a_{K+1} + 2(K+2)(K+1) a_{K+2} - a_K] x^K \dots = 0$$

$$\Rightarrow \begin{cases} 4a_2 - a_0 = 0 \\ (k+1)ka_{k+1} + 2(k+2)(k+1)a_{k+2} - a_k = 0 \\ \text{for } k=1, 2, 3, \dots \end{cases}$$

Solve for a_2 and a_{k+2} to get:

$$\begin{cases} a_2 = a_0/4 \\ a_{k+2} = \frac{a_k - k(k+1)a_{k+1}}{2(k+2)(k+1)} \quad k=1, 2, 3, \dots \end{cases}$$

We need two LI power series solutions y_1, y_2 .

y_1

$$a_0 = 1$$

$$a_1 = 0$$

 y_2

$$a_0 = 0$$

$$a_1 = 1$$

$$a_2 = \frac{a_0}{4} = \frac{1}{4}$$

$$\boxed{k=1}$$

$$a_3 = \frac{a_1 - 2a_2}{2(3)(2)} = -\frac{1}{24}$$

$$\boxed{k=2}$$

$$a_4 = \frac{a_2 - 6a_3}{2(4)(3)} = \frac{1}{48}$$

$$a_2 = \frac{a_0}{4} = 0$$

$$a_3 = \frac{a_1 - 2a_2}{12} = \frac{1}{12}$$

$$a_4 = \frac{a_2 - 6a_3}{24} = -\frac{1}{48}$$

The two LI solutions are:

$$\begin{cases} y_1 = 1 + \frac{x^2}{4} - \frac{x^3}{24} + \frac{x^4}{48} + \dots \\ y_2 = x + \frac{x^3}{12} - \frac{x^4}{48} + \dots \end{cases}$$

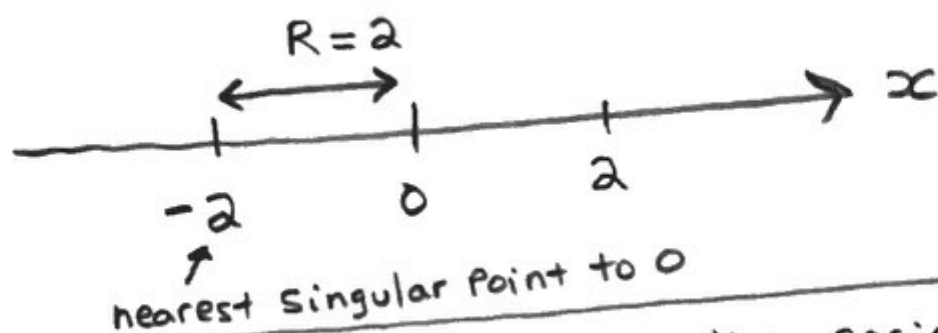
The general solution is

$$y = c_1 y_1 + c_2 y_2$$

$$y = c_1 \left(1 + \frac{x^2}{4} - \frac{x^3}{24} + \frac{x^4}{48} + \dots \right) + c_2 \left(x + \frac{x^3}{12} - \frac{x^4}{48} + \dots \right)$$

The DE $(x+2)y'' - y = 0$

has a singular point at $x = -2$.



The radius of convergence of the series is then at least $R=2$.

The series solutions converge at least

for $\boxed{-2 < x < 2}$.

In the general case, to solve

$$y'' + P(x)y' + Q(x)y = 0$$

about the ordinary point $x_0 = 0$,

We expand $P(x)$ and $Q(x)$ as
power series (Maclaurin series)

$$\left\{ \begin{array}{l} P(x) = p_0 + p_1 x + p_2 x^2 + \dots \\ Q(x) = q_0 + q_1 x + q_2 x^2 + \dots \end{array} \right\} \rightarrow \text{Since } P(x) \text{ and } Q(x) \text{ are analytic at } 0, \text{ the series exist.}$$

We then have to multiply series
to deal with $P(x)y'$ and $Q(x)y$.

The next example will illustrate this.

ex: (7) Find the first five terms of a Power series solution of the

$$\text{IVP} \quad \begin{cases} y'' + e^{-x} y = 0 \\ y(0) = 3 \\ y'(0) = -2 \end{cases}$$

Since we only want five terms, let's look for $y = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + \dots$
We will not use sigma notation for this example.

The Maclaurin series of e^x is

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

if we replace x by $(-x)$, then:

$$e^{-x} = 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \dots$$

$$y = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + \dots$$

$$\Rightarrow y' = a_1 + 2a_2 x + 3a_3 x^2 + 4a_4 x^3 + \dots$$

$$\Rightarrow y'' = 2a_2 + 6a_3 x + 12a_4 x^2 + \dots$$

Plug the series in $y'' + e^{-x} y = 0$.

$$\underbrace{(2a_2 + 6a_3 x + 12a_4 x^2 + \dots)}_{y''} + \underbrace{\left(1 - x + \frac{x^2}{2!} - \dots\right)}_{e^{-x}} \underbrace{(a_0 + a_1 x + a_2 x^2 + \dots)}_y$$

$$\dots = 0$$

Multiply the series of e^{-x}
and the series of y .

$$y'' = 2a_2 + 6a_3x + 12a_4x^2 + \dots$$

$$\begin{aligned} e^{-x}y &= \left(1 - x + \frac{x^2}{2!} - \dots\right) (a_0 + a_1x + a_2x^2 + \dots) \\ &= a_0 + (a_1x - a_0x) + \left(a_2x^2 - a_1x^2 + \frac{a_0x^2}{2}\right) + \dots \end{aligned}$$

$$y'' + e^{-x}y = 0$$

$$(2a_2 + a_0) + (6a_3 + a_1 - a_0)x + \left(12a_4 + a_2 - a_1 + \frac{a_0}{2}\right)x^2 + \dots$$

Set all coefficients of x^k to 0.

$$\begin{cases} 2a_2 + a_0 = 0 \\ 6a_3 + a_1 - a_0 = 0 \\ 12a_4 + a_2 - a_1 + \frac{a_0}{2} = 0 \end{cases}$$

Solve for a_2, a_3, a_4 .

$$\begin{cases} a_2 = -\frac{a_0}{2} \\ a_3 = \frac{a_0 - a_1}{6} \\ a_4 = \frac{-\frac{a_0}{2} + a_1 - a_2}{12} \end{cases}$$

The initial conditions give a_0 and a_1 .

$$\begin{cases} a_0 = y(0) = 3 \\ a_1 = y'(0) = -2 \end{cases}$$

$$a_2 = -3/2$$

$$a_3 = 5/6$$

$$a_4 = \frac{-3/2 - 2 + 3/2}{12} = -1/6$$

The Power series solution of the IVP
is then:

$$y = 3 - 2x - \frac{3}{2}x^2 + \frac{5}{6}x^3 - \frac{1}{6}x^4 + \dots$$

Since the DE has no singular
Point, the series converges
for all x .
